

Basic Undefined Terms and Geometric Notation

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A **point** represents a location and has no size or shape.

A **line** extends infinitely in two opposite directions.

A **ray** starts at one endpoint and extends infinitely in one direction.

A **line segment** has two endpoints and a definite length.

An **angle** is formed by two rays with a common endpoint called the vertex.

A **plane** is a flat surface that extends infinitely in all directions.

Geometric Notation

Geometric figures are represented using standard **geometric notations**.

Point

A point can be named using a capital letter.

Point *A*

Line

A line can be named using two points on the line or a lowercase letter.

$$\overleftrightarrow{AB}$$

or

line l

Ray

A ray can be named using its endpoint first, followed by another point on the ray.

$$\overrightarrow{AB}$$

Line Segment

A line segment can be named using its two endpoints.

$$\overline{AB}$$

Angle

An angle can be named using three points, with the vertex written in the middle.

$$\angle ABC$$

Plane

A plane can be named using a capital letter or three noncollinear points.

Plane P

or

plane ABC

Examples

Example 1

Write the correct geometric notation for a point named M .

Point M

Example 2

Write the correct geometric notation for a line passing through points A and B .

$$\overleftrightarrow{AB}$$

Example 3

Write the correct geometric notation for a ray that starts at point C and passes through point D .

$$\overrightarrow{CD}$$

Example 4

Write the correct geometric notation for a line segment with endpoints E and F .

$$\overline{EF}$$

Example 5

Write the correct geometric notation for an angle with vertex at G and sides passing through points H and I .

$$\angle HGI$$

Example 6

Write the correct geometric notation for a plane named R .

Plane R

Practice Exercises

Practice 1

Write the correct geometric notation for a point named M .

Practice 2

Write the correct geometric notation for a line passing through points A and B .

Practice 3

Write the correct geometric notation for a ray that starts at point C and passes through point D .

Practice 4

Write the correct geometric notation for a line segment with endpoints E and F .

Practice 5

Write the correct geometric notation for an angle with vertex at G and sides passing through points H and I .

Practice 6

Write the correct geometric notation for a plane named R .

Practice 7

Write the correct geometric notation for a line passing through points J and K .

Practice 8

Write the correct geometric notation for a ray that starts at point L and passes through point M .

Practice 9

Write the correct geometric notation for a line segment with endpoints N and O .

Practice 10

Write the correct geometric notation for a plane determined by points P , Q , and R .

Activities

Activity 1

Write the correct geometric notation for a point named T .

Activity 2

Write the correct geometric notation for a line passing through points C and D .

Activity 3

Write the correct geometric notation for a ray that starts at point E and passes through point F .

Activity 4

Write the correct geometric notation for a line segment with endpoints G and H .

Activity 5

Write the correct geometric notation for an angle with vertex at J and sides passing through points K and L .

Activity 6

Write the correct geometric notation for a plane named S .

Activity 7

Write the correct geometric notation for a line passing through points M and N .

Activity 8

Write the correct geometric notation for a ray that starts at point P and passes through point Q .

Activity 9

Write the correct geometric notation for a line segment with endpoints R and S .

Activity 10

Write the correct geometric notation for a plane determined by points U , V , and W .

Thank you!